

Seismic demand analysis of steel buildings with masonry infill walls using machine learning-based time-history surrogate model

Mao Li and Yong Li*

Department of Civil & Environmental Engineering, University of Alberta, Edmonton T6H2G7, Canada

ABSTRACT: Masonry infill walls (MIWs) affect the seismic performance of steel frame buildings differently depending on the seismic hazard level, yet their structural contribution is often oversimplified or neglected in building design, under the assumption that doing so is conservative. Consequently, seismic simulation and performance evaluation of frame building frames often use bare frames due to the complexity of modeling the highly nonlinear behavior masonry infill walls and their interactions with surrounding beams and columns. Integrating MIWs into Performance-Based Earthquake Engineering (PBEE) frameworks for masonry infill frames requires high-fidelity finite element models capable of capturing wall-frame interactions, which exacerbates the computational burden of probabilistic seismic risk assessment. To address this bottleneck, this study introduces a physics-informed surrogate model integrating Temporal Convolutional Networks with Bidirectional Gated Recurrent Units (*PhyTCN-BiGRU*) to predict seismic response time histories based on earthquake ground excitations. This architecture is specifically designed to map ground motion input time-histories to multiple structural response output time-histories while capturing the complex hysteretic degradation associated with infilled frames. Validated on an eight-story steel moment-resisting frame with MIWs, the *PhyTCN-BiGRU* framework accurately predicts nonlinear responses, including residual drifts, with a 14-fold increase in computational speed. This approach offers a scalable pathway for rigorous seismic risk analysis that fully accounts for the structural influence of infill walls without prohibitive computational costs.

1. INTRODUCTION

Nature hazards, such as earthquake, cause millions of economic loss and casualties worldwide, underscoring the critical need for probabilistic frameworks, such as Performance-Based Earthquake Engineering (PBEE), to guide engineering decisions to mitigate seismic loss and enhance resilience (Li et al. 2024; Shi et al. 2025; Zhou et al. 2021). To be effective, PBEE requires accurate probabilistic assessments of structural performance under frequent to rare seismic events. However, a significant challenge in modern seismic design is accurately accounting for non-structural components that participate in the structural response, particularly Masonry Infill Walls (MIWs).

While often treated as architectural elements, MIWs interact significantly with the primary bounding frame, altering the building's stiffness, strength, and energy dissipation capacity. Although they can provide additional lateral resistance, MIWs are also prone to brittle failure modes and can induce adverse effects such as soft-story mechanisms or short-column effects. Accurately capturing these behaviors requires extensive Nonlinear Time-History Analysis (NLTHA) using high-fidelity Finite Element (FE) models with sophisticated hysteretic materials (e.g., pinching and strength degradation). This imposes a massive computational burden that often forces structural engineers to simplify MIW models or neglect their

structural contributions entirely, potentially leading to inaccurate seismic risk estimates (Günay and Mosalam 2013).

As an alternative to rigorous MIW modeling for computational efficiency, Machine Learning (ML) has emerged as a powerful tool for surrogate model development. Surrogate models, such as data-driven Deep Neural Networks (DNN) (Zeng et al. 2025) or Physics-Informed Neural Networks (PINN) (Zhang et al. 2020), have shown promise in accelerating structural response prediction. Despite their potential, existing studies often struggle to accurately predict the highly nonlinear responses of complex systems across the full spectrum of seismic intensities, particularly regarding permanent residual drifts when the structure undergoes significant inelastic deformation (Chen et al. 2025).

To overcome the limitations of existing neural networks for structural response prediction, this study proposes a Physics-Informed Temporal Convolutional Network enhanced with Bidirectional Gated Recurrent Units (*PhyTCN-BiGRU*). The architecture leverages two distinct components: a TCN that utilizes dilated convolutions to efficiently extract local, high-frequency features of time series while mitigating the vanishing gradient problem in long sequences, and a BiGRU that captures the global, long-term trends necessary to model the effect of hysteretic energy dissipation and permanent deformation. This synergistic design ensures both accuracy and efficiency, enabling the *PhyTCN-BiGRU* to serve as a robust alternative to traditional NLTHA for seismic response time-history prediction.

By deploying this pre-trained surrogate model, structural demands can be simulated rapidly when subjected to earthquake ground motions, significantly accelerating seismic risk assessment. In this study, the proposed framework is trained and validated on an 8-story steel frame building with MIWs, followed by a comprehensive seismic demand and risk analysis. The paper is organized as follows: Section 2 details the methodology for surrogate model development, high-fidelity FE model used for data generation, and seismic risk analysis; Section 3 presents the validation results and the seismic demand assessment; and Section 4 summarizes the conclusions.

2. 2026-03-17 9:04:00 AM METHODOLOGY

2.1 Framework of seismic risk analysis using machine learning-based time-history surrogate model

2.1.1 Introduction of the framework

To accelerate the seismic risk assessment, this study proposes a new neural network architecture. The proposed architecture uses a Temporal Convolutional Network (TCN) integrated with Bidirectional Gated Recurrent Units (BiGRU), which is trained using time-history data generated from high-fidelity structural models under ground motion accelerations (Li and Li 2026). Furthermore, to incorporate underlying physics into the training process, the total loss function is augmented with additional terms that enforce physics-based constraints derived from the governing equation. Without loss of generality, the dynamics of a building system with nonlinear restoring forces subjected to earthquake excitations are governed by the equation of motion (EoM) shown in Equation [1]:

$$[1] \mathbf{M}\ddot{\mathbf{u}}(t) + \mathbf{C}\dot{\mathbf{u}}(t) + \mathbf{F}_s[\mathbf{u}(t), \boldsymbol{\theta}] = -\mathbf{M}\boldsymbol{\Gamma}a_g(t)$$

Here, $\ddot{\mathbf{u}}(t)$, $\dot{\mathbf{u}}(t)$, and $\mathbf{u}(t)$ correspond to the acceleration, velocity, and displacement vectors at a given time t , respectively. $\mathbf{M}$ and $\mathbf{C}$ denote the mass and damping matrices of the system, respectively. The nonlinear restoring force vector $\mathbf{F}_s$ is a function of the displacement vector $\mathbf{u}(t)$ and the hysteretic parameters $\boldsymbol{\theta}$. The term $a_g(t)$ indicates ground motion acceleration at time t , while the load influence vector $\boldsymbol{\Gamma}$ defines the manner in which this acceleration is applied to each DOF within the structural model. Normalizing Equation (1) by the mass matrix yields Equation [2].

$$[2] \ddot{\mathbf{u}}(t) + \mathbf{M}^{-1}\mathbf{C}\dot{\mathbf{u}}(t) + \mathbf{M}^{-1}\mathbf{F}_s[\mathbf{u}(t), \boldsymbol{\theta}] = -\boldsymbol{\Gamma}a_g(t)$$

Reorganizing this further, one can rewrite Equation [2] into Equation [3] by collecting the two mass-related terms.

$$[3] \ddot{\mathbf{u}}(t) + \mathbf{f}_g[\dot{\mathbf{u}}(t), \mathbf{u}(t), \boldsymbol{\theta}] = -\mathbf{\Gamma} \mathbf{a}_g(t)$$

Here, the mass-normalized internal force is $\mathbf{f}_g[\dot{\mathbf{u}}(t), \mathbf{u}(t), \boldsymbol{\theta}] = \mathbf{M}^{-1} \mathbf{C} \dot{\mathbf{u}}(t) + \mathbf{M}^{-1} \mathbf{F}_s[\mathbf{u}(t), \boldsymbol{\theta}]$. The objective is to construct a surrogate model $\mathbf{Z}(t) = \mathcal{N}(\mathbf{a}_g(t); \boldsymbol{\theta})$ that can predict the structural response time-histories with high accuracy at a fraction of the computational cost, where $\boldsymbol{\theta}$ is the hyper-parameter vector, and $\mathbf{Z}(t) = \{\mathbf{z}_1(t), \mathbf{z}_2(t), \mathbf{z}_3(t)\}^T = \{\mathbf{u}(t), \dot{\mathbf{u}}(t), \mathbf{f}_g[\dot{\mathbf{u}}(t), \mathbf{u}(t), \boldsymbol{\theta}]\}^T$. To this end, this study uses *PhyTCN-BiGRU* to bridge the ground motion acceleration time-history ($\mathbf{a}_g(t)$) to the structural time-history response ($\mathbf{Z}(t)$). The overall framework integrates finite element software (e.g., *OpenSees*) for high-fidelity seismic response data generation, the machine learning platform (e.g., *TensorFlow*) for *PhyTCN-BiGRU* model construction, physics-informed loss function definition, and an iteration process for model training, response inference for seismic demand and risk analysis, as schematically shown in Figure 1.

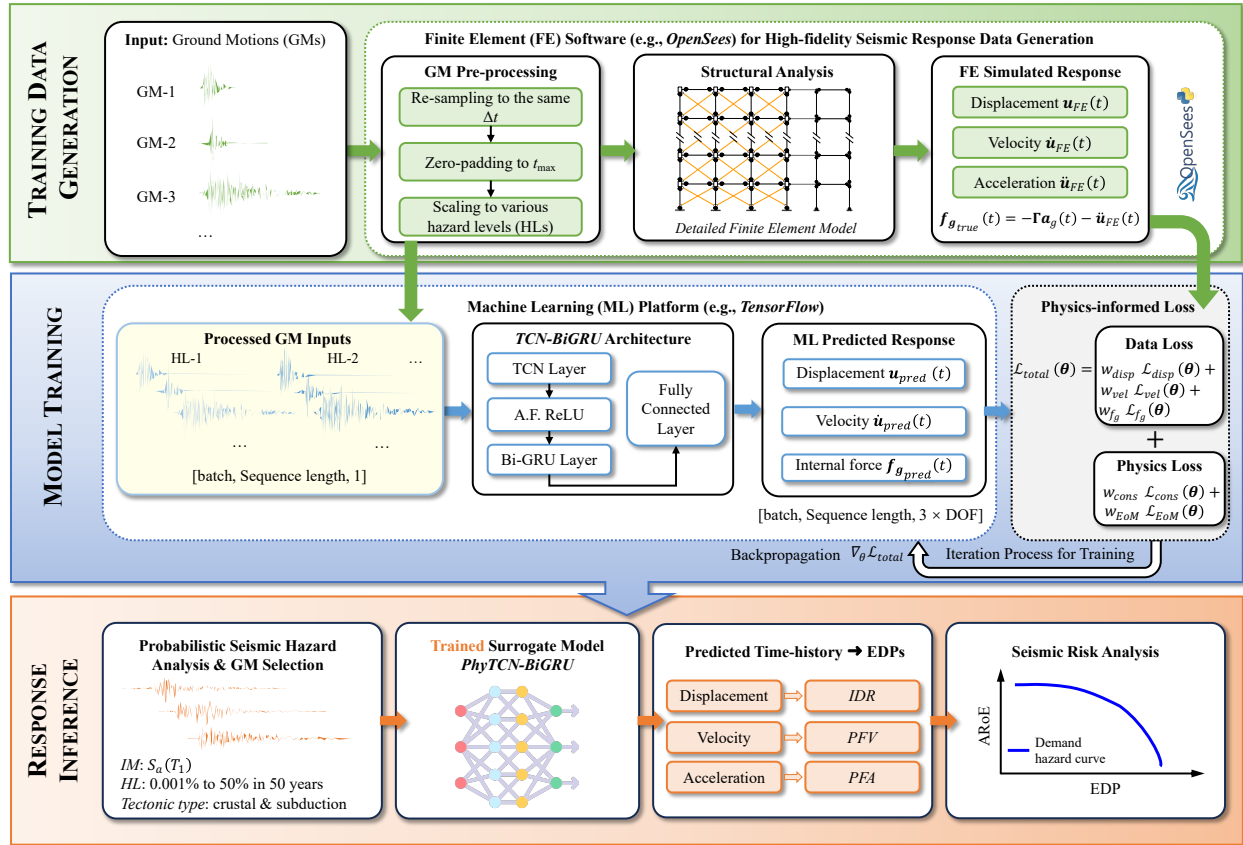


Figure 1: Schematic view of proposed framework

2.1.2 Physics-informed and data loss functions

The loss function is a critical component for the *PhyTCN-BiGRU* network training. In this study, a hybrid loss function that combines both data-driven and physics-informed components is adopted, as shown in Equation [4].

$$[4] \mathcal{L}_{total}(\boldsymbol{\theta}) = w_{disp} \mathcal{L}_{disp}(\boldsymbol{\theta}) + w_{vel} \mathcal{L}_{vel}(\boldsymbol{\theta}) + w_{fg} \mathcal{L}_{fg}(\boldsymbol{\theta}) + w_{cons} \mathcal{L}_{cons}(\boldsymbol{\theta}) + w_{EoM} \mathcal{L}_{EoM}(\boldsymbol{\theta})$$

where the total loss consists of data losses and physics-informed losses. $\mathcal{L}_{disp}(\boldsymbol{\theta})$, $\mathcal{L}_{vel}(\boldsymbol{\theta})$, and $\mathcal{L}_{fg}(\boldsymbol{\theta})$ represent losses arising from the inaccuracy of the ML-predicted displacement, velocity, and mass-normalized internal force histories, respectively, compared to the FE-simulated ones. These contribute to the data loss, as defined in Equations [5] to [7], which quantify discrepancies between the predicted outputs (i.e., displacement $\mathbf{z}_1$, velocity $\mathbf{z}_2$, mass-normalized internal force $\mathbf{z}_3$) and FE-simulated responses (i.e., u_{FE} , $\dot{u}_{FE}$, and $f_{g,FE}$) at all timesteps.

$$[5] \mathcal{L}_{disp}(\boldsymbol{\theta}) = \sum_{i=1}^n \left[\left\| \mathbf{z}_1^{(i)}(\boldsymbol{\theta}) - \mathbf{u}_{FE}^{(i)} \right\|_2^2 \right]$$

$$[6] \mathcal{L}_{vel}(\boldsymbol{\theta}) = \sum_{i=1}^n \left[\left\| \mathbf{z}_2^{(i)}(\boldsymbol{\theta}) - \dot{\mathbf{u}}_{FE}^{(i)} \right\|_2^2 \right]$$

$$[7] \mathcal{L}_{fg}(\boldsymbol{\theta}) = \sum_{i=1}^n \left[\left\| \mathbf{z}_3^{(i)}(\boldsymbol{\theta}) - \mathbf{f}_{g,FE}^{(i)} \right\|_2^2 \right]$$

Here, i is the time index and n is the length of the time-series sequence.

The physics-informed loss includes the consistency loss $\mathcal{L}_{cons}(\boldsymbol{\theta})$ and the EoM loss $\mathcal{L}_{EoM}(\boldsymbol{\theta})$. The consistency loss aims to measure how the derivatives of the predicted displacement histories match the predicted velocity histories, as shown in Equation [8].

$$[8] \mathcal{L}_{cons}(\boldsymbol{\theta}) = \sum_{i=1}^n \left[\left\| \dot{\mathbf{z}}_1^{(i)}(\boldsymbol{\theta}) - \mathbf{z}_2^{(i)}(\boldsymbol{\theta}) \right\|_2^2 \right]$$

The EoM loss aims to measure the residual of the governing equation of motion, as defined in Equation [9].

$$[9] \mathcal{L}_{EoM}(\boldsymbol{\theta}) = \sum_{i=1}^n \left[\left\| \ddot{\mathbf{z}}_2^{(i)}(\boldsymbol{\theta}) - \mathbf{z}_3^{(i)}(\boldsymbol{\theta}) + \mathbf{r}a_g^{(i)} \right\|_2^2 \right]$$

The coefficient w_i represents the weights. Finally, the network is optimized by minimizing $\mathcal{L}_{total}(\boldsymbol{\theta})$ using standard optimizers Adaptive moment estimation (Adam) (Kingma and Ba 2017).

2.2 High-fidelity finite element model of steel frame building with masonry infill walls

To generate training data based on the high-fidelity finite element (FE) model, an 8-story SMRF buildings are designed in accordance with Nation Building Code of Canada (Canadian Commission on Building and Fire Codes 2022) and Canadian Standards S16:24 (CSA Group 2024). A symmetric rectangular plan view was adopted, as shown in Figure 2. The design-related information of SMRF and MIW are presented in Table 1.

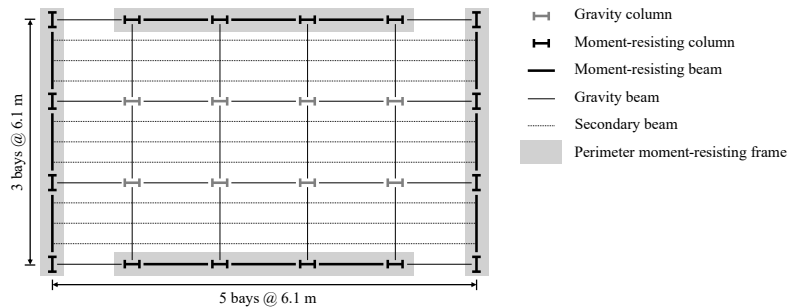


Figure 2: Plan view of archetype buildings

Table 1: Design-related information of steel frame building

	Parameter	Value
SMRF	Construction location	City Hall, Vancouver, BC, Canada
	V_{s30}	450 m/s
	Frame type	Type-D moment-resisting frame
	Story height	4.0 m (4.6 m for first floor)
	Steel material	ASTM A992 Gr. 50
MIW	Prism elastic modulus	1980 MPa
	Prism strength	1.9 MPa
	Panel thickness	125 mm
	Infill configuration	Fully infilled in SMRF

The FE model of 8-story steel frame building with MIWs was developed by Python version of Open System for Earthquake Engineering Simulation (OpenSeesPy) (Zhu et al. 2018), as shown in Figure 3. The FE model consists of two major components: the moment-resisting frame, which captures the nonlinear lateral-force-resisting behavior, and an equivalent gravity frame, which accounts for the P-Δ effects (Elkady and Lignos 2015). Beams and columns within the moment-resisting frame are modelled by concentrated plasticity methods and *IMK* material (Lignos and Krawinkler 2011; Lignos et al. 2019). The nonlinear behavior of panel zone areas is also considered using a rectangular configuration with a rotational spring modelled by *Hysteretic* material (Skiadopoulos et al. 2021). The equivalent diagonal struts method is utilized to model the behavior of MIWs. The compression only *Pinching4* material is assigned to struts, whose parameters are (Huang et al. 2020).

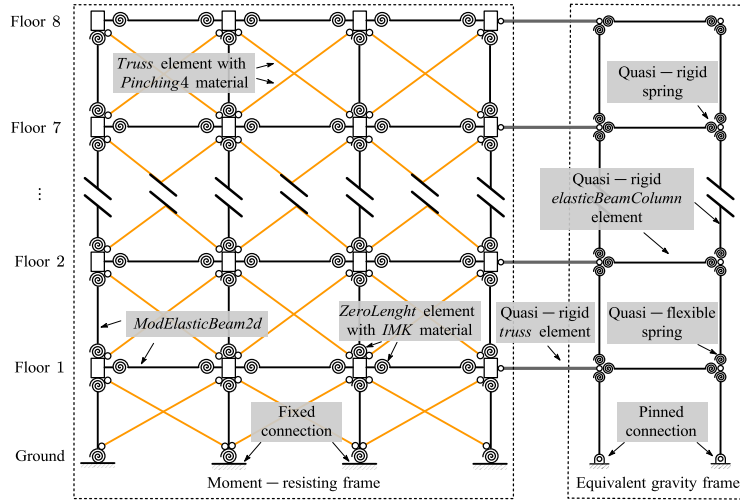


Figure 3: FE model of 8-story steel frame building with MIWs

2.3 Seismic risk analysis

The seismic risk is quantified by the annual rate of exceedance (ARoE) of certain EDP at intensity measures (IMs), $\lambda(edp)$, as shown in Equation [10].

$$[10] \lambda(edp) = \int P(EDP > edp|im) d\lambda(im)$$

where $\lambda(im)$ is the seismic hazard curve (the ARoE of given im), $P(EDP > edp|im)$ denotes the PoE a given EDP at im level, which can be derived by Equation [11].

$$[11] P(EDP > edp|im) = 1 - \Phi\left(\frac{\ln(edp) - \mu_{\ln EDP}(im)}{\beta_{\ln EDP}(im)}\right)$$

where $\Phi(\cdot)$ is the standard normal cumulative distribution function. $\mu_{\ln EDP}(im)$ and $\beta_{\ln EDP}(im)$ represent the logarithmic mean and dispersion of EDP at given im , respectively. To calculate the seismic demand, the piecewise interpolation and extrapolation method proposed by Bradley et al. (Bradley 2013) is adopted in current study to calibrate $\mu_{\ln EDP}(im)$ and $\beta_{\ln EDP}(im)$, respectively.

3. RESULTS

3.1 Surrogate model training

The model was trained using 22 far-fault ground motions recommended by FEMA P-695 (Applied Technology Council 2009), scaled to 20 seismic hazard levels ranging from 0.01% to 50% probability of exceedance in 50 years. This resulted in a dataset of 440 ground motion simulations. The training process utilized the hyperparameters summarized in Table 2.

Table 2: Hyper-parameters in the surrogate model training

Parameter		Value
TCN	Number of filters	128
	Kernel size	5
	Number of stacks	2
	Dilations	1, 2, 4, 8, 16, 32, 64
	Dropout rate	5%
BiGRU	Number of units	128
Model training	Batch size	8
	Epoch	2500
	Learning rate	0.001
	Data split	Training : validation = 8:2
	CPU	Intel i7-9700K 3.6GHz
	GPU	NVIDIA GTX 1080 8GB

To evaluate the performance of the surrogate model, Figure 4 compares the FE-simulated and *PhyTCN-BiGRU* predicted roof time-history responses under three representative seismic hazard levels: Design-Based Earthquake (DBE, 10% in 50 years), Maximum Considered Earthquake (MCE, 2% in 50 years), and Extreme Rare Earthquake (ERE, 0.1% in 50 years). The *PhyTCN-BiGRU* predictions exhibit high correlation with the FE results across all seismic intensities. Notably, the model demonstrates exceptional capability in capturing permanent residual deformations at high seismic hazard levels (e.g., ERE level), a critical requirement for accurate PBEE applications.

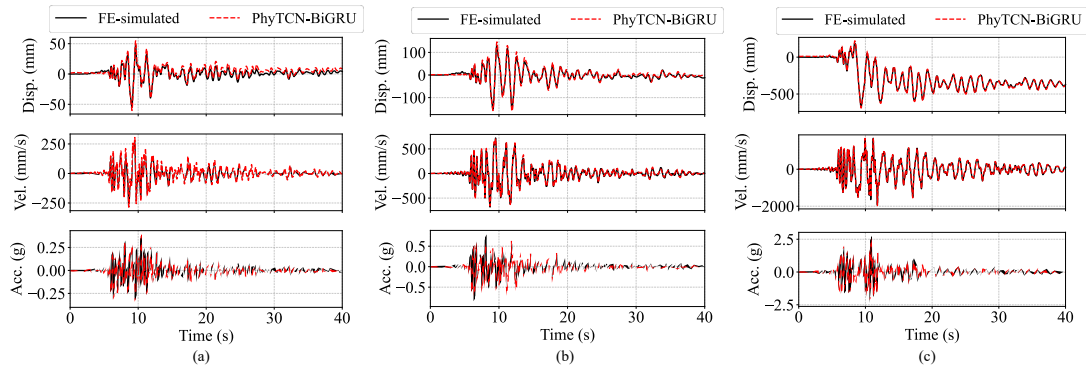


Figure 4 Roof time-history responses at (a) DBE, (b) MCE, and (c) ERE

To further quantify the surrogate model prediction accuracy, Figure 5 compares three critical Engineering Demand Parameters (EDPs), Peak Floor Displacement (PFD), Peak Floor Velocity (PFV), Peak Floor Acceleration (PFA), and Residual floor displacement (RFD) between the FE simulation and the surrogate model. The high determination coefficients ($R^2 > 0.98$) observed for both training and validation datasets confirm the model's robustness and feasibility for practical application.

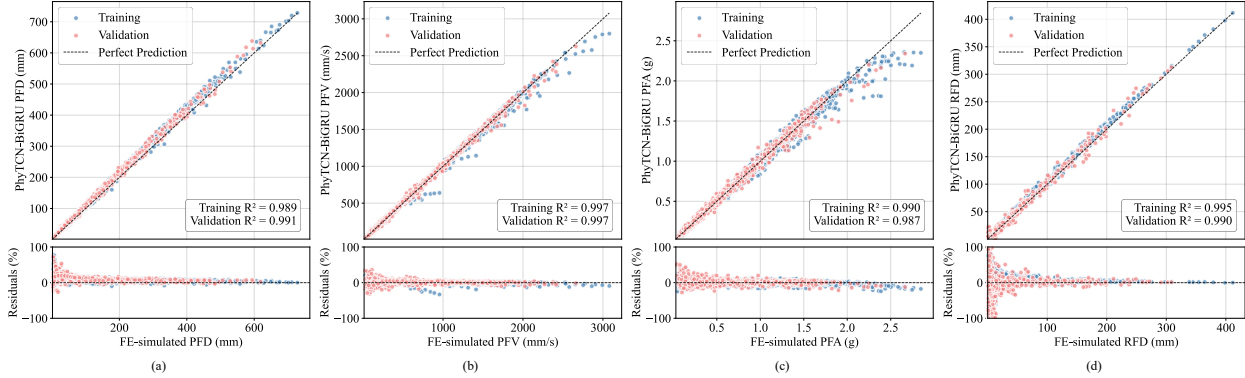


Figure 5 EDPs comparison between FE-simulation and ML-prediction: (a) PFD, (b) PFV, (c) PFA, and (d) RFD.

3.2 Seismic risk analysis

Prior to risk analysis, a probabilistic seismic hazard analysis (PSHA) and ground motion selection were conducted (Li and Li 2025). To account for uncertainty, 500 ground motions were selected across 16 hazard levels (0.1% to 0.5% in 50 years), totaling 8,000 simulations. Figure 6 illustrates the predicted distributions for Maximum Inter-story Drift Ratio (MIDR) and Maximum Peak Floor Acceleration (MPFA).

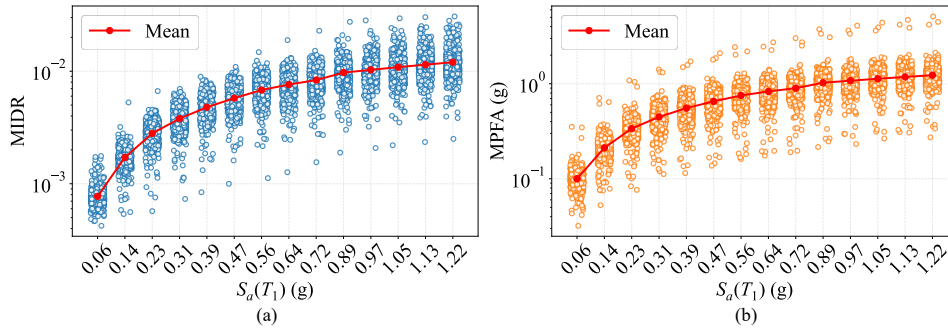


Figure 6 Predicted (a) MIDR and (b) MPFA at 16 hazard levels

Using Equation [10], the seismic hazard curve was convolved with the conditional demand probability to derive the demand hazard curve, shown in Figure 7.

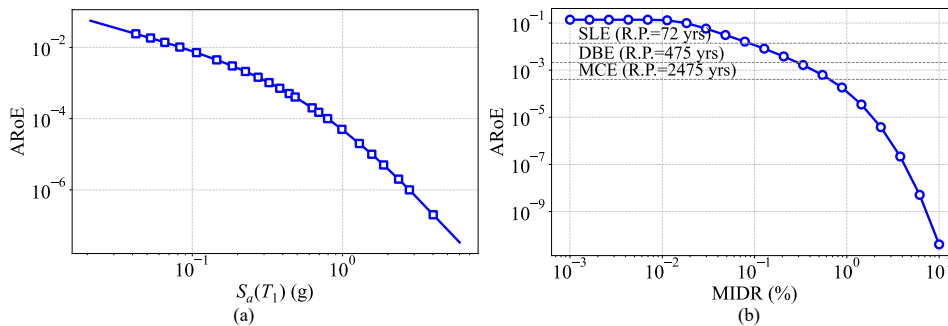


Figure 7 Hazard curves: (a) seismic hazard curve, (b) demand hazard curve

A primary advantage of the proposed surrogate model is its computational efficiency. Table 3 compares the computational cost of the FE simulation versus the *PhyTCN-BiGRU* model. The surrogate model significantly reduces inference time per ground motion, achieving an overall speed-up ratio of 14.0 for 8,000 simulations. This demonstrates the substantial efficiency gains achievable through the proposed framework

Table 3: Computational efficiency and speed-up comparison between FE simulation and PhyTCN–BiGRU

Time	FE simulation (OpenSeesPy)	PhyTCN-BiGRU
One-time training time	N/A	~1413 mins
Inference time (per GM)	~ 15.2 mins*	0.07 sec
Total time for 8,000 runs	121,600 mins	8,661 mins (440×15.2+1,413+0.07×8,000)
Speed-up ratio	1.0	14.0

*The inference time is calculated based on the average simulation time of 440 training and validation data.

4. CONCLUSIONS

This study presented a rapid seismic risk assessment framework utilizing a physics-informed surrogate model (*PhyTCN-BiGRU*), specifically tailored to address the computational challenges of analyzing steel frames with masonry infill walls (MIWs). The integration of MIWs significantly increases the complexity of numerical models, often creating a bottleneck for probabilistic performance-based design. The proposed *PhyTCN-BiGRU* architecture overcomes this by synergizing Temporal Convolutional Networks (TCN) for local feature extraction with Bidirectional Gated Recurrent Units (BiGRU) for global trend modeling. This hybrid design effectively captures the complex, path-dependent hysteretic behaviors inherent in strut-frame interactions, enabling accurate prediction of nonlinear responses across a full spectrum of seismic intensities.

Validated on a high-fidelity 8-story infilled steel frame, the surrogate model demonstrated exceptional accuracy, particularly in capturing the critical permanent residual deformations often exacerbated by infill damage. Furthermore, the framework achieved a speed-up ratio of 14.0 compared to traditional finite element simulations, reducing the total analysis time for comprehensive risk assessment from months to days. These findings confirm that the *PhyTCN-BiGRU* approach provides a robust solution for the rigorous inclusion of MIWs in Performance-Based Earthquake Engineering. By removing the computational barriers associated with high-fidelity modeling, this framework paves the way for more realistic seismic risk assessments that fully account for the structural contributions of non-structural partitions.

ACKNOWLEDGMENTS

The authors acknowledge the financial support from the Natural Sciences and Engineering Research Council (NSERC) of Canada through the Discovery Grants Program (RGPIN2024-05989) and the China Scholarship Council.

REFERENCES

- Applied Technology Council. 2009. *Quantification of Building Seismic Performance Factors*. Washington, D.C., USA.
- Bradley, B. A. 2013. "Practice-oriented estimation of the seismic demand hazard using ground motions at few intensity levels." *Earthquake Engineering & Structural Dynamics*, 42 (14): 2167–2185. <https://doi.org/10.1002/eqe.2319>.
- Canadian Commission on Building and Fire Codes. 2022. *National Building Code of Canada: 2020*. National Research Council of Canada.

- Chen, J., Y. Shi, and Y. Yu. 2025. "A C-PINN method for structural seismic response modeling." *Structures*, 82: 110422. <https://doi.org/10.1016/j.istruc.2025.110422>.
- CSA Group. 2024. *CSA S16:24 Design of steel structures*. Toronto, ON, Canada: Canadian Standards Association.
- Elkady, A., and D. G. Lignos. 2015. "Effect of gravity framing on the overstrength and collapse capacity of steel frame buildings with perimeter special moment frames." *Earthquake Engineering & Structural Dynamics*, 44 (8): 1289–1307. <https://doi.org/10.1002/eqe.2519>.
- Günay, S., and K. M. Mosalam. 2013. "PEER Performance-Based Earthquake Engineering Methodology, Revisited." *Journal of Earthquake Engineering*, 17 (6): 829–858. <https://doi.org/10.1080/13632469.2013.787377>.
- Huang, H., H. V. Burton, and S. Sattar. 2020. "Development and Utilization of a Database of Infilled Frame Experiments for Numerical Modeling." *Journal of Structural Engineering*, 146 (6): 04020079. American Society of Civil Engineers. [https://doi.org/10.1061/\(ASCE\)ST.1943-541X.0002608](https://doi.org/10.1061/(ASCE)ST.1943-541X.0002608).
- Kingma, D. P., and J. Ba. 2017. "Adam: A Method for Stochastic Optimization." arXiv.
- Li, M., and Y. Li. 2025. "Probabilistic performance evaluation of steel moment resisting frame with masonry infill walls." In: *14th International Conference on Structural Safety and Reliability - ICOSSAR'25*. Los Angeles.
- Li, M., and Y. Li. 2026. "Physics-informed Temporal Convolutional Network with Bidirectional Gated Recurrent Unit developed for rapid prediction of seismic response time-histories of building structures." *Computer-Aided Civil and Infrastructure Engineering*, 100008. <https://doi.org/10.1016/j.cacaie.2026.100008>.
- Li, M., M. Shi, and W. Fu. 2024. "A variable damping viscous damper for control of buildings under wind loading." *Journal of Building Engineering*, 97: 110960. <https://doi.org/10.1016/j.jobbe.2024.110960>.
- Lignos, D. G., A. R. Hartloper, A. Elkady, G. G. Deierlein, and R. Hamburger. 2019. "Proposed Updates to the ASCE 41 Nonlinear Modeling Parameters for Wide-Flange Steel Columns in Support of Performance-Based Seismic Engineering." *Journal of Structural Engineering*, 145 (9): 04019083. American Society of Civil Engineers. [https://doi.org/10.1061/\(ASCE\)ST.1943-541X.0002353](https://doi.org/10.1061/(ASCE)ST.1943-541X.0002353).
- Lignos, D. G., and H. Krawinkler. 2011. "Deterioration Modeling of Steel Components in Support of Collapse Prediction of Steel Moment Frames under Earthquake Loading." *Journal of Structural Engineering*, 137 (11): 1291–1302. American Society of Civil Engineers. [https://doi.org/10.1061/\(ASCE\)ST.1943-541X.0000376](https://doi.org/10.1061/(ASCE)ST.1943-541X.0000376).
- Shi, M., W. Fu, M. Li, Z. Li, and H. Wang. 2025. "Seismic performance evaluation of isolated liquid storage tanks with passive variable damping and stiffness devices." *Mechanical Systems and Signal Processing*, 241: 113523. <https://doi.org/10.1016/j.ymssp.2025.113523>.
- Skiadopoulos, A., A. Elkady, and D. G. Lignos. 2021. "Proposed Panel Zone Model for Seismic Design of Steel Moment-Resisting Frames." *Journal of Structural Engineering*, 147 (4): 04021006. American Society of Civil Engineers. [https://doi.org/10.1061/\(ASCE\)ST.1943-541X.0002935](https://doi.org/10.1061/(ASCE)ST.1943-541X.0002935).
- Zeng, C., W. Guo, S. Zhang, C. He, P. Shao, Y. Sun, and P. Liao. 2025. "Recursive DeepONet for surrogate modelling of nonlinear dynamics and its application in real-time hybrid simulation." *Engineering Structures*, 338: 120621. <https://doi.org/10.1016/j.engstruct.2025.120621>.
- Zhang, R., Y. Liu, and H. Sun. 2020. "Physics-informed multi-LSTM networks for metamodeling of nonlinear structures." *Computer Methods in Applied Mechanics and Engineering*, 369: 113226. <https://doi.org/10.1016/j.cma.2020.113226>.
- Zhou, W., S. Guo, X. Deng, and D. Xu. 2021. "Livelihood resilience and strategies of rural residents of earthquake-threatened areas in Sichuan Province, China." *Natural Hazards*, 106 (1): 255–275. Springer Netherlands. <https://doi.org/10.1007/s11069-020-04460-4>.
- Zhu, M., F. McKenna, and M. H. Scott. 2018. "OpenSeesPy: Python library for the OpenSees finite element framework." *SoftwareX*, 7: 6–11. <https://doi.org/10.1016/j.softx.2017.10.009>.